

RAG for Speech-Language Pathology

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Introduction

Traumatic brain injury (TBI) rehabilitation is a complex, interdisciplinary process that demands constant access to the latest neurological research. Therapists must manually sift through a vast and ever-growing body of scientific literature, creating a significant lag between discovery and clinical application that ultimately limits patient outcomes.

Equally critical, TBI recovery involves non-professional family members and caretakers who need to understand the patient's condition in plain, accessible language. This group is frequently left without meaningful information, increasing caretaker stress and slowing recovery at home.

This project addresses both challenges with a **Retrieval-Augmented Generation (RAG)** system, an AI tool that makes TBI research rapidly accessible to clinicians, therapists, and family caretakers alike, bridging the gap between medical expertise and everyday caregiving.

Questions

- Can a RAG system effectively surface relevant TBI research for both clinical and non-clinical users?
- Does integrating patient-specific context via physician's notes improve the relevance and usefulness of AI-generated responses?
- Can the system meaningfully close the knowledge gap between medical professionals and family caretakers of TBI patients?

Methods

Using **ChromaDB** as a vector database and **DeepSeek** as the large language model backbone, a full RAG pipeline was built in **Python**. Neurological research literature was ingested and indexed into ChromaDB. When a user submits a query, the system performs semantic retrieval of the most relevant documents, which are passed alongside the query to DeepSeek to generate a grounded, readable response.



A key feature developed following initial caretaker feedback allows users to optionally attach a **physician's note** or clinical summary to their query. This document is incorporated directly into the retrieval context, allowing the system to produce responses tailored to the specific patient's situation without requiring model fine-tuning. Users retain full control over how much clinical detail they share, preserving privacy and maintaining compliance with relevant health data policies.

The completed system was evaluated in a structured feedback session with a patient's caretaker. The participant queried the system on topics directly relevant to their real-world caregiving situation and provided detailed feedback on usability, information relevance, and the clarity of outputs for a non-clinical audience. This feedback directly shaped the personalization feature, establishing an iterative loop between end-user need and system design.

The literature review underpinning this project identified three core challenges motivating the system: the high degree of clinical uncertainty inherent in TBI outcomes, the persistent lag between new research and clinical application, and the widely reported information gap experienced by family caretakers. RAG architectures were selected specifically for their ability to ground AI responses in verified source documents, reducing hallucination and increasing the trustworthiness of AI-generated clinical guidance.

Results

FINDING 1 - USABILITY

The caretaker found the system highly usable and effective for quickly accessing information about the patient's TBI condition. Non-professional users without clinical training were able to retrieve meaningful, readable content with minimal friction, demonstrating that the system's accessibility extends well beyond a clinical-only audience and into everyday caregiving contexts.

FINDING 2 - INFORMATION RELEVANCE

While system responses were consistently accurate and grounded in the indexed research literature, the caretaker noted that outputs could be more tailored to the specific patient's situation. General TBI guidance, while useful as a starting point, was sometimes less actionable than desired given the nuances of their unique caregiving context. This finding highlighted personalization as the primary area for system improvement.

FINDING 3 - PERSONALIZATION AS SOLUTION

The relevance gap identified in Finding 2 directly motivated the development of the optional physician's note upload feature. By allowing caretakers to share relevant clinical details about the patient, the system was able to incorporate this context into the retrieval pipeline and produce noticeably more relevant, specific, and actionable responses. Critically, this personalization was achieved entirely within the RAG framework without fine-tuning the underlying model, and users retained full control over what information they chose to share, preserving both privacy and regulatory compliance.

FINDING 4 - DUAL-AUDIENCE VALUE

The system demonstrated particular value for non-professional caretakers, a population typically underserved by clinical AI tools, which are most often designed with healthcare providers as the primary user. The ability to receive clear, grounded, plain-language answers to complex TBI questions validated the dual-audience design as both feasible and impactful. This positions the system as a novel contribution to AI-assisted rehabilitation: a tool that serves the full ecosystem of care around a TBI patient, not just the clinical team.

Discussion

Results support the feasibility of a RAG-based system for improving TBI information access across both clinical and non-clinical stakeholders. Grounding responses in indexed peer-reviewed literature represents a meaningful advance over manual literature search for therapists, and a significant accessibility improvement for family caretakers navigating a complex and often deeply uncertain recovery process.

The optional physician's note feature proved to be a critical design insight: meaningful personalization is achievable within a RAG framework without fine-tuning, and user-controlled information sharing preserves both privacy and policy compliance. Importantly, this feature emerged directly from end-user feedback, underscoring the value of iterative, user-centered design in clinical AI development.

The dual-audience approach, simultaneously serving clinicians seeking research synthesis and family members seeking plain-language guidance, may offer a replicable model for AI-assisted health communication beyond TBI rehabilitation, applicable wherever complex medical knowledge must be made accessible across a spectrum of users with varying levels of clinical expertise.

Future Directions

Custom model training: Replace DeepSeek with a domain-specific model trained exclusively on neurological and TBI literature to improve response accuracy, reduce third-party API dependence, and increase system autonomy, contingent on access to greater computational resources than currently available.

Expanded literature base: Continuously ingest newly published peer-reviewed articles into ChromaDB to improve the system's breadth and currency of knowledge, ensuring responses remain aligned with the evolving state of neurological research.

Automated web scraping pipeline: Implement an automated article scraping system with rigorous quality filtering to ensure only credible, peer-reviewed sources are indexed, keeping the knowledge base current without manual curation overhead.

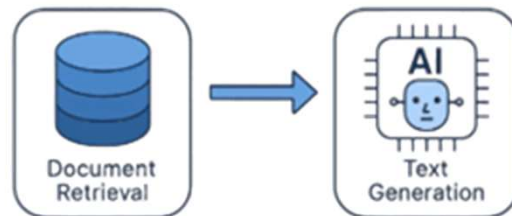
Broader pilot study: Expand evaluation beyond a single caretaker to include multiple caretakers and therapists across varied TBI cases and clinical settings, strengthening the validity of findings and surfacing additional usability improvements for future system iterations.

References

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[Additional references provided upon request]

What is RAG in AI?



The hybrid model RAG (Retrieval-Augmented Generation) bridges retrieval systems and generative models to generate responses.